

Optimal Approximation for Fully Dynamic Matching and Ordered Ruzsa–Szemerédi Graphs: A Feasibility-Checked Execution Plan for the Deterministic and Lower-Bound Frontiers

Research plan prepared for Ruifeng Cao

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Abstract

Behnezhad and Ghafari (FOCS 2024) and Assadi, Khanna, and Kiss (SODA 2025) reduced fully dynamic $(1 - \varepsilon)$ -approximate maximum matching to a purely combinatorial quantity, the density $\text{ORS}(n)$ of *ordered Ruzsa–Szemerédi graphs*: the current state of the art maintains the *edges* of a $(1 - \varepsilon)$ -approximate maximum matching in amortized $n^{o(1)} \cdot \text{ORS}(n)$ update time, with high probability, against an adaptive adversary. Two questions remain open: **(OP-D)** can a *deterministic* algorithm achieve the same guarantees, and **(OP-L)** can current lower-bound techniques (hidden permutations, RS-graph reductions, OMv-based hardness) be extended to prove polynomial lower bounds against *deterministic* dynamic algorithms?

This document is an execution plan for attacking both questions. It contains: (i) a verified map of the 2024–2026 state of the art, including Pratt’s SODA 2026 proof that ordered and classical RS densities are polynomially equivalent, which collapses part of the design space; (ii) precise formalizations of OP-D and OP-L, including a correction of a common misreading of the SODA 2025 result (it improved the *update time*, not the approximation ratio); (iii) several original structural observations, among them a fully proved $\Theta(n^2)$ -vs- $\tilde{O}(n)$ deterministic/randomized separation for matching-size estimation in the adjacency-probe model, and a derived *architectural barrier* showing that the entire randomized polylog-update pipeline (BKSW / Behnezhad / ABR / BG / AKK) has no deterministic drop-in replacement at its sublinear-oracle layer; (iv) an honest taxonomy of what kinds of deterministic lower bounds are achievable today versus blocked by known cell-probe barriers; and (v) four concrete workstreams with tasks, decision gates, fallback publishable units, a 9–12 month timeline, a risk register, and a claim-status appendix. All citations were re-verified against arXiv/DOI meta-data in June 2026 except where explicitly tagged.

Contents

1	Scope, reading of the open problem, and one correction	2
1.1	The open problem as posed	2
1.2	A correction to the premise [V]	3
1.3	What changed in 2025–2026 and why it reshapes the plan [V]	3
1.4	Deliverables of this plan	4

2	Verified state of the art (June 2026)	4
2.1	Model zoo: what exactly is being maintained	4
2.2	Ordered Ruzsa–Szemerédi graphs [M]	4
2.3	Upper bounds for fully dynamic matching: verified table	5
2.4	Randomness audit of the ORS pipeline [V]	6
2.5	Lower-bound landscape (verified) and where determinism currently plays no role .	7
3	Formal problems and original structural observations	8
3.1	The two problems, formalized	8
3.2	Why the existing techniques cannot be “extended unchanged”: a no-go observation [P]	8
3.3	A composition observation linking ORS to OMv [S]	9
3.4	A proved deterministic/randomized separation in the probe model, and the architectural barrier [P]	9
3.5	Consistency checks that calibrate difficulty [P]	10
4	Workstream A: derandomization (OP-D)	11
5	Workstream B: the combinatorics of ORS/RS density	12
6	Workstream C: deterministic-specific lower bounds (OP-L)	13
7	Workstream D: engineering, verification, packaging	14
8	Phase 0 verification, timeline, gates, fallbacks	14
8.1	Timeline (9–12 months) and decision gates	14
8.2	Fallback publishable units (pre-committed)	15
9	Risk register	15
A	Randomness audit table (Task A0, v0.3)	15
B	Claim-status appendix	21
C	Notation	22

1 Scope, reading of the open problem, and one correction

1.1 The open problem as posed

The prompt poses, in the context of “optimal approximation for fully dynamic matching and ordered Ruzsa–Szemerédi (ORS) graphs”:

- (a) *Can deterministic algorithms achieve the same approximation guarantees as the randomized ORS-based algorithms of Behnezhad–Ghafari [BG24] and Assadi–Khanna–Kiss [AKK25]?*
- (b) *Can current lower-bound techniques (hidden permutations, RS graphs) be extended to prove polynomial communication lower bounds for deterministic algorithms in the dynamic setting?*

We formalize these as Problems 3.1 (OP-D) and 3.2 (OP-L) in Section 3, after fixing the model zoo in Section 2; the distinctions between maintaining *edges*, maintaining a *size estimate*, and maintaining a *witness* turn out to be essential to both questions, and several natural readings of (a) and (b) are either already resolved or provably hopeless with current technology, so the formalization step is not bureaucratic.

1.2 A correction to the premise [V]

The prompt states that Assadi–Khanna–Kiss (SODA 2025) “used ordered RS graphs to significantly improve the *approximation ratio* of fully dynamic matching.” The verified record is slightly different, and the difference matters for what OP-D should ask:

- Behnezhad–Ghafari [BG24] (FOCS 2024, pp. 314–327; arXiv:2404.06069) already achieve approximation ratio $1 - \varepsilon$ for every fixed $\varepsilon > 0$. They introduce ORS graphs and maintain the *edges* of a $(1 - \varepsilon)$ -approximate maximum matching in amortized update time $O(\sqrt{n^{1+O(\varepsilon)} \cdot \text{ORS}(n)})$, with high probability, against an adaptive adversary.
- Assadi–Khanna–Kiss [AKK25] (SODA 2025, pp. 2971–2990; arXiv:2406.13573) keep the same approximation ratio $1 - \varepsilon$ and the same adaptive-adversary guarantee but improve the *update time* to amortized $n^{o(1)} \cdot \text{ORS}(n)$, thereby reducing the dynamic problem *entirely* to bounding the combinatorial quantity $\text{ORS}(n)$.

So the “optimal approximation” in the title of this research direction is $1 - \varepsilon$, which both papers already attain (randomized); the open algorithmic frontier is (i) the update time as a function of $\text{ORS}(n)$, (ii) the truth about $\text{ORS}(n)$ itself, and (iii) determinism. OP-D below is phrased accordingly.

1.3 What changed in 2025–2026 and why it reshapes the plan [V]

Two verified developments postdate the original formulation of this open problem and materially change the attack surface:

1. **Pratt (SOSA 2026): ordered $\approx$ unordered.** Pratt [Pra26] (arXiv:2502.02455, DOI 10.1137/1.9781611978964.27) proves that if $\text{ORS}(n) \geq \Omega(n^c)$ for a fixed $c > 0$, then for every fixed $\delta > 0$ the *classical* RS density at slightly smaller (still linear) matching size satisfies $\text{RS}(n, \Theta_\delta(n)) \geq \Omega(n^{c(1-\delta)})$. Consequently the ordering relaxation does *not* buy polynomial density, the question “is $\text{ORS}(n) = n^{o(1)}$ or $n^{1-o(1)}$?” from [BG24] reduces (in the polynomial regime) to the decades-old Ruzsa–Szemerédi question, and the AKK update time is, up to $n^{o(1)}$, governed by *classical* RS density. For this plan, the consequence is that Workstream B (Section 5) should target RS and ORS jointly, and that “find a denser *ordered* construction” is no longer a promising shortcut to faster algorithms. (Exact parameter translation: Phase-0 Task 14.)
2. **Bernstein–Bhattacharya–Kiss–Saranurak (STOC 2025): deterministic maximal matching in sublinear time.** [BBKS25] (arXiv:2504.20780, DOI 10.1145/3717823.3718153) give a deterministic fully dynamic *maximal* matching in amortized $\tilde{O}(n^{8/9})$ update time, breaking the long-standing $O(n)$ deterministic barrier on dense graphs, plus a randomized adaptive-adversary algorithm in $\tilde{O}(n^{3/4})$. This confirms that (i) the deterministic frontier

is active and the community considers it major, and (ii) even *maximal* matching — a $1/2$ -approximation primitive — currently costs deterministic $n^{8/9}$, which calibrates how far OP-D is from resolution and supplies the natural first stepping stone (target D1 below).

1.4 Deliverables of this plan

The plan is organized as: verified state of the art (Section 2); formal problems, original observations, and a proved probe-model separation with its barrier corollary (Section 3); four workstreams WS-A–WS-D with concrete tasks (Sections 4–7); timeline, decision gates, and fallback publishable units (Section 8); risk register (Section 9); claim-status and notation appendices (Appendices B–C).

2 Verified state of the art (June 2026)

2.1 Model zoo: what exactly is being maintained

Let $G = (V, E)$, $|V| = n$, $|E| = m$, undergo edge insertions and deletions; $\mu(G)$ denotes the maximum matching size. The literature mixes four distinct objectives, and conflating them produces false “contradictions” between upper and lower bounds. We fix names:

EDGES(ε): maintain an explicit edge set $M \subseteq E$ that is a matching with $|M| \geq (1 - \varepsilon)\mu(G)$ after every update.

SIZE(α) / **SIZE**($\alpha, \beta n$): maintain a number $\hat{\mu}$ with $\alpha\mu(G) - \beta n \leq \hat{\mu} \leq \mu(G)$ (multiplicative α , optional additive slack βn).

FRAC(α): maintain a fractional matching of value at least $\alpha\mu(G)$ (or within α of the fractional optimum).

WITNESS-VC(α): maintain an explicit vertex set that is an α -approximate vertex cover.

Orthogonal axes: *amortized vs. worst-case* update time; *oblivious vs. adaptive* adversary; *randomized (whp) vs. deterministic*. One important calibration [V]: the ORS-based algorithms [BG24, AKK25] are already robust to *adaptive* adversaries, so OP-D is a pure determinism question — the usual intermediate target “upgrade oblivious to adaptive” is unavailable as a stepping stone; the remaining gap between [AKK25] and OP-D is exactly “whp $\rightarrow$ always-correct, no coins.”

2.2 Ordered Ruzsa–Szemerédi graphs [V]

Definition 2.1 (RS and ORS graphs; [RS78, BG24]). A graph G on n vertices is an RS *graph with parameters* (r, t) if its edge set can be partitioned into t edge-disjoint *induced* matchings $M_1, \dots, M_t$, each of size r . It is an *ordered RS (ORS) graph with parameters* (r, t) if there is an ordering $M_1, \dots, M_t$ of t edge-disjoint matchings of size r such that every M_i is an induced matching in the subgraph $M_i \cup M_{i+1} \cup \dots \cup M_t$. Write $\text{RS}(n, r)$, $\text{ORS}(n, r)$ for the maximum such t , and $\text{RS}(n)$, $\text{ORS}(n)$ for the regime $r = \Theta(n)$ (constant locality hidden in the Θ , as in [BG24]).

Every RS graph is an ORS graph under any ordering, so $\text{ORS}(n, r) \geq \text{RS}(n, r)$. Verified facts:

- *Lower bound (constructions)*. $\text{RS}(\cdot, \Theta(n)) \geq n^{1/\Theta(\log \log n)}$, via Fischer, Lehman, Newman, Raskhodnikova, Rubinfeld, and Samorodnitsky [FLNRRS02] (attribution per [BG24]); this is super-polylogarithmic but $n^{o(1)}$. Behrend-set-based constructions [Beh46, RS78] give $n^{1-o(1)}$ induced matchings of size $n^{1-o(1)}$ (slightly sublinear r). [V] (regime details: verify exact constants in Task 14).

- *Upper bounds.* For $r = \Theta(n)$ no upper bound of the form $n^{1-\Omega(1)}$ is known: whether $\text{ORS}(n) = n^{o(1)}$ or $\text{ORS}(n) = n^{1-o(1)}$ is wide open [BG24]. For r very close to $n/2$ (e.g. $r \geq (1/4 + \delta)n$ and stronger regimes) classical triangle-removal/regularity arguments force t to be small; the precise (r, t) frontier is scattered across the literature and is collected as Task 7. [M] for the near- $n/2$ regime.
- *Equivalence.* Pratt [Pra26]: polynomial ORS density implies polynomial RS density at linear matching size (Section 1.3); so in the polynomial regime, ordered and unordered densities coincide up to n^δ losses. [V]

2.3 Upper bounds for fully dynamic matching: verified table

Throughout, “det” = deterministic, “rand” = randomized (whp), “adp” = correct against adaptive adversaries, “obl” = oblivious adversary assumed. All rows verified against the cited source unless tagged [M].

(a) EDGES(ε), i.e. $(1 - \varepsilon)$ -approximate matching maintained explicitly.

Update time (amortized unless noted)	Type	Source	Notes
$O(\sqrt{m}\varepsilon^{-2})$ (worst-case)	det	[GP13]	FOCS’13, pp. 548–557
$n/(\log^* n)^{\Omega(1)}$	rand	[ABKL23]	regularity-based [M] (pages)
$n/2^{\Omega(\sqrt{\log n})}$ (bipartite)	rand	[Liu24]+[LW17]	via online matrix-vector
$O(\sqrt{n^{1+O(\varepsilon)}\text{ORS}(n)})$	rand, adp	[BG24]	FOCS’24, pp. 314–327
$n^{o(1)} \cdot \text{ORS}(n)$	rand, adp	[AKK25]	SODA’25, pp. 2971–2990

Calibration [V]: for approximation ratios below $2/3$ (equivalently $\varepsilon > 1/3$), update time $n^{1/2+o(1)}$ -type bounds follow from Bernstein–Stein [BS16] ($3/2 + \varepsilon$ in $O(m^{1/4})$); for $\varepsilon \leq 1/3$ the best unconditional bound not parameterized by ORS is the $n/2^{\Omega(\sqrt{\log n})}$ of [Liu24] (this calibration is stated in [Pra26]).

(b) Maximal matching (a canonical $1/2$ -approximation primitive).

Update time	Type	Source	Notes
$O(\sqrt{m})$ (worst-case)	det	[NS13]	arXiv:1207.1277
$\tilde{O}(n^{8/9})$	det	[BBKS25]	STOC’25; first $o(n)$ det on dense G
$O(\log n)$	rand, obl	[BGS11]	SICOMP’18 corrected version [M]
$O(1)$	rand, obl	[Sol16]	FOCS’16, pp. 325–334
$\tilde{O}(n^{3/4})$	rand, adp	[BBKS25]	

(c) SIZE estimation and better-than- $1/2$ ratios in polylog time (all randomized).

Guarantee	Update time	Source	Notes
$\text{SIZE}(1/2 + \Omega(1))$	polylog	[Beh23]	SODA’23, pp. 129–162
$\text{SIZE}(1 + 1/\sqrt{2} - \varepsilon)^{-1}$ bip., 1.973^{-1} gen.	polylog wc, adp	[BKSW23]	SODA’23 + JACM
$\text{SIZE}(2 - \sqrt{2} - \varepsilon)$ general	polylog	[ABR24]	SODA’24, pp. 3040–3061
$\text{SIZE}(1 - \varepsilon)$	$m^{1/2 - \Omega_\varepsilon(1)}$	[BKS23]	FOCS’23, arXiv:2302.05030

Here $2 - \sqrt{2} \approx 0.5857$; [ABR24] also push to $(2 - \sqrt{2} - \varepsilon)$ for any fixed $\varepsilon > 0$ in polylog time. [V]

(d) Deterministic landscape beyond maximal.

Guarantee	Update time	Source	Notes
EDGES($1/2 - \varepsilon$) (<i>i.e.</i> $(2+\varepsilon)$ -apx)	$\text{poly}(\log n, 1/\varepsilon)$	[BHN16]	STOC'16, arXiv:1604.05765
better-than-2 apx, bipartite	$O(n^\gamma)$, any $\gamma > 0$	[BHN16]	second result of the paper
$(3/2 + \varepsilon)$ -apx	$O(m^{1/4}\varepsilon^{-2.5})$	[BS16]	SODA'16, pp. 692–711
$(2 + \varepsilon)$ -apx FRAC+VC	$O(1)$	[BCH17]	IPCO'17
det rounding of dyn. fractional	polylog overhead	[BK21]	ICALP'21 [M]
EDCS maintenance, worst-case	$O(m^{1/4})$ -type	[GSSU22]	SOSA'22, pp. 12–23 [M]

Two verified summary points. (i) No deterministic algorithm is known with polylog (or even $n^{o(1)}$) update time for *any* constant-factor guarantee better than $(2 + \varepsilon)$, in any of the four objectives; the deterministic better-than-2 frontier is $O(n^\gamma)/O(m^{1/4})$ -type [BHN16, BS16]. (ii) No deterministic algorithm with update time $o(\sqrt{m})$ is known for EDGES(ε) at small ε ; even matching Liu’s randomized $n/2^{\Omega(\sqrt{\log n})}$ deterministically is open, because the underlying OMv algorithm of Larsen–Williams [LW17] (SODA'17, pp. 2182–2189) is randomized (determinism status of subpolynomial-saving OMv: verify in Task 14). [M]/[M]

2.4 Randomness audit of the ORS pipeline [M]

Where exactly do coins enter [BG24, AKK25] and the polylog SIZE algorithms? This audit drives Workstream A.

Update (v1.1). Task A0’s read-through passes 1–2 are complete; the machine-readable audit table (v0.3, lemma-level locations, auto-generated by `audit_lab`) is reproduced in Appendix A, with the full technical discussion in the companion A0 note. Three sharpenings of the prose audit below [M]: (i) BG24’s coins live *only* in its Lemma 7 sampling plus the estimator/sparsification interfaces — the ORS charging (Lemma 11), certificate bookkeeping, and McGregor boosting (derandomized by Tirodkar, per AKK25 Prop. 2.2) are deterministic; (ii) the derandomization problem therefore reduces to exactly three modules (opportunistic sampler; Beh21 estimator interface; vertex sparsification); (iii) ABR24’s Theorem 2 is a *deterministic* two-pass streaming $(2 - \sqrt{2} - \varepsilon)$ -approximation, so the combinatorial core is deterministic under pass access and the dynamic coins are forced by the probe access model alone.

1. *Sublinear-time estimators as inner oracles.* The polylog-update pipelines [BKSW23, Beh23, ABR24] repeatedly invoke sublinear-time matching/size estimators — most prominently random greedy maximal matching (RGMM) estimation in the style of Behnezhad [Beh21] (FOCS'21, pp. 873–884) — on implicitly represented (kernel/contracted/peeled) graphs accessed through adjacency probes. These estimators are Monte Carlo: per call they probe a sublinear number of adjacency entries and succeed whp.
2. *Random ranks/permutations.* RGMM correctness and locality analyses are driven by a uniformly random permutation of edges/vertices; adaptive-adversary robustness in [BKSW23, AKK25] is obtained by re-randomizing and by outputting only low-sensitivity quantities (sizes), plus standard amplification.
3. *Sampling for sparsification/peeling.* [BG24, AKK25] use random vertex/edge sampling inside the ORS-driven phase structure to bound the number of “epochs” by ORS density; failure probabilities are union-bounded over epochs.

Consequence (formalized as Corollary 3.8): the pipeline’s randomness is concentrated in a clean interface — “estimate μ of an implicitly represented graph from few probes” — and that

interface is *provably* impossible to implement deterministically in sublinear probes. Derandomizing the pipeline therefore cannot be a local substitution; it must change the architecture.

2.5 Lower-bound landscape (verified) and where determinism currently plays no role

- *OMv-conditional*. Henzinger–Krinninger–Nanongkai–Saranurak [HKNS15] (STOC’15, pp. 21–30): under the OMv conjecture, many exact dynamic problems need $n^{1-o(1)}$ update time. Liu [Liu24] (FOCS’24, pp. 228–243) proves *equivalences*: maintaining a $(1 + \varepsilon)$ -approximate WITNESS-VC in amortized $n^{1-c}\varepsilon^{-C}$ time is possible iff OMv is false; maintaining $(1 - \varepsilon)$ -approximate bipartite matching in n^{1-c} iff a nontrivial “dynamic approximate OMv” algorithm exists. By König duality the *value* of the minimum vertex cover equals μ in bipartite graphs, and value maintenance *is* sublinearly possible [BKS23]; the hardness is for *witness* maintenance — one more reason the model zoo of Section 2.1 matters. [V] (exact quantifiers: Task 14.)
- *Static sublinear (query) lower bounds, randomized*. Behnezhad–Roghani–Rubinstein [BRR24] (STOC’24, arXiv:2406.08595): for every $\delta > 0$ there is $\varepsilon > 0$ such that estimating μ within additive εn requires $\Omega(n^{2-\delta})$ adjacency-list queries; near-matching upper bounds in $n^{2-\Omega_\varepsilon(1)}$ [BKS23, BRRS23]. These already hold for randomized algorithms, so they cannot, by themselves, separate det from rand. [M]
- *Hidden permutations (the technique named in the prompt)*. Assadi–Sundaresan [AS23] (FOCS’23, pp. 909–932; arXiv:2310.05728) build *permutation-hiding* graphs directly on RS graphs and prove that $(1 - \varepsilon)$ -approximating bipartite matching in semi-streaming needs $\Omega(\log(1/\varepsilon)/\log(1/\beta))$ passes, where n^β is the RS density available to the construction. This is a *distributional* (hence randomized-robust) communication argument in a *bounded-space* model. [M]
- *Dynamic streams*. Assadi–Khanna–Li–Yaroslavtsev [AKLY16] show $n^{2-3\varepsilon}$ -space lower bounds for n^ε -approximation in dynamic streams — again randomized-robust, and so strong already for randomized algorithms that there is little deterministic/randomized daylight left in that model. [M]
- *Cell-probe ceilings*. Unconditional cell-probe lower bounds for dynamic data structures top out around $\Omega((\log n / \log \log n)^2)$ (Larsen [Lar12]); polynomial unconditional bounds are a notorious open frontier (Pătraşcu’s multiphase program [Pat10], nondeterministic barriers [CGL15]). Nothing here exploits determinism, and no known route gives polynomial *deterministic-only* cell-probe bounds either. [M]

Reading for OP-L. Every polynomial lower bound in this landscape is either conditional (OMv-style), or lives in a restricted information model (streams, communication, query/probe), and *all* of them are randomized-robust. There is, verifiably, no published polynomial lower bound anywhere in the dynamic-matching literature that applies to deterministic algorithms but not to randomized ones. That gap is exactly the opportunity — and Section 3 shows it cannot be closed by “extending” the existing techniques unchanged.

3 Formal problems and original structural observations

3.1 The two problems, formalized

Problem 3.1 (OP-D: determinism). Design a *deterministic* fully dynamic algorithm that maintains the edges of a $(1 - \varepsilon)$ -approximate maximum matching with amortized update time $f(\varepsilon) \cdot \text{ORS}(n) \cdot n^{o(1)}$, i.e. matching [AKK25] up to $n^{o(1)}$ factors — or refute the possibility in a meaningful restricted model. Graded sub-targets, each independently publishable:

- D1 (maximal/half).** Deterministic fully dynamic *maximal* matching in $n^{o(1)}$ (ultimately polylog) update time; intermediate: beat the $\tilde{O}(n^{8/9})$ of [BBKS25].
- D2 (fractional).** Deterministic $(1 + \varepsilon)$ -approximate *fractional* matching in polylog update time (current det: $(2 + \varepsilon)$ in $O(1)$ [BCH17]).
- D3 (size, better-than-2).** Derandomize the polylog-time $\text{SIZE}(1/2 + \Omega(1))$ algorithms [Beh23, BKS23, ABR24] — any constant ratio $> 1/2$ in deterministic $n^{o(1)}$ time.
- D4 (ORS-parameterized).** The full OP-D statement above.

Problem 3.2 (OP-L: deterministic lower bounds). Prove super-polylogarithmic (ideally polynomial) lower bounds on the update/query complexity of *deterministic* fully dynamic matching algorithms, in a model strong enough to be meaningful, by extending or replacing the hidden-permutation/RS-reduction/OMv toolkits. Taxonomy of admissible answer types (used throughout WS-C):

- L1.** Unconditional deterministic lower bounds in oracle/probe models (achievable now; see Lemma 3.6).
- L2.** Architectural barriers: theorems that the *existing algorithmic frameworks* cannot be derandomized (achievable now; see Corollary 3.8).
- L3.** Conditional bounds under a new deterministic hardness hypothesis (“Det-OMv”), with deterministic reductions.
- L4.** ORS/RS *density lower bounds* (constructions): these lower-bound the runtime of the leading algorithm family [BG24, AKK25], not of all algorithms — a legitimate but weaker currency.
- L5.** Extensions of permutation hiding [AS23] to dynamic settings with *bounded space or bounded recourse*; we prove below why some such restriction is forced.

3.2 Why the existing techniques cannot be “extended unchanged”: a no-go observation [P]

Observation 3.3 (Randomized-robust techniques cannot give deterministic-only polynomial bounds at ratios below $2 - \sqrt{2}$). Call a lower-bound technique *randomized-robust* if any bound it proves for deterministic algorithms it also proves (up to constants) for randomized ones — as is the case for distributional/communication arguments via Yao’s principle (hidden permutations [AS23], RS-reduction streaming bounds, [BRR24]) and for OMv-style reductions, whose source hypothesis is itself about randomized algorithms [HKNS15]. Then no randomized-robust technique can prove a polynomial update-time lower bound for deterministic $\text{SIZE}(2 - \sqrt{2} - \delta)$ maintenance for any $\delta > 0$: such a bound would equally apply to randomized algorithms and contradict the verified polylog-time randomized algorithm of [ABR24]. The same argument with [BKS23, Beh23] blocks all ratios in $(1/2, 2 - \sqrt{2}]$ and with [BBKS25] blocks $\omega(n^{3/4+o(1)})$ bounds for adaptive randomized maximal matching.

Consequences. (i) A deterministic polynomial lower bound at ratio better than $1/2 + \delta$ must use determinism *essentially* — e.g. adversary arguments that adapt to the algorithm’s fixed behaviour, fooling-set/rank-style deterministic communication bounds, or a deterministic hardness hypothesis (L3). (ii) For ratios in $(2 - \sqrt{2}, 1)$, polynomial lower bounds are open *even for randomized* algorithms (Liu [Liu24] covers the $\varepsilon \rightarrow 0$ end only conditionally/by equivalence), so OP-L at those ratios subsumes a major open randomized question; the honest deterministic-specific opportunity is at ratios where randomized algorithms are already fast, i.e. $[1/2, 2 - \sqrt{2}]$, and in the parameterized ORS regime.

3.3 A composition observation linking ORS to OMv [S]

Observation 3.4 (If ORS is sparse, dynamic approximate OMv is easy). Suppose $\text{ORS}(n) = n^{o(1)}$. Then [AKK25] maintains $\text{EDGES}(\varepsilon)$ in amortized $n^{o(1)}$ time; feeding this into Liu’s equivalence [Liu24] (algorithm direction) yields a nontrivial “dynamic approximate OMv” algorithm in the sense of [Liu24]. Contrapositive: hardness of dynamic approximate OMv would imply $\text{ORS}(n) = n^{\Omega(1)}$, i.e. a *conditional combinatorial lower bound* for RS-type densities — an unusual direction (complexity $\Rightarrow$ extremal combinatorics) worth a short note if the parameters compose. Status: sketch; the quantifier matching (ε dependence, value-vs-edges, bipartiteness) is exactly Phase-0 Task 14(c).

3.4 A proved deterministic/randomized separation in the probe model, and the architectural barrier [P]

The following is elementary but, to our knowledge after the June 2026 verification pass, not stated in this form in the dynamic-matching literature; its role here is to convert the randomness audit of Section 2.4 into a theorem-grade barrier and to seed Workstream C. (Closest precedent: deterministic-vs-randomized gaps for *learning* a hidden matching [ABKRS04], a harder reconstruction problem.)

Definition 3.5 (Adjacency-probe model). An algorithm accesses an unknown graph G on vertex set $[n]$ only via probes $\text{ADJ}(u, v) \in \{0, 1\}$, chosen adaptively. Cost = number of probes.

Lemma 3.6 (Quadratic deterministic probe bound for gap matching). *Any deterministic adaptive algorithm that distinguishes $\mu(G) = 0$ from $\mu(G) \geq n/4$ must make more than $\binom{\lceil n/2 \rceil}{2} - 1 = \Omega(n^2)$ probes in the worst case. Consequently no deterministic algorithm with $o(n^2)$ probes can estimate $\mu(G)$ within additive $n/8$, nor within any multiplicative factor with additive slack $< n/8$. In contrast, the same gap problem is solvable by a randomized algorithm with $O(n \log n)$ probes (error probability $1/\text{poly}(n)$), and $\Omega(n)$ probes are necessary for randomized algorithms; so the deterministic and randomized probe complexities of the gap problem are $\Theta(n^2)$ and $\tilde{\Theta}(n)$ respectively — a quadratic separation.*

Proof. Deterministic lower bound. Run the deterministic algorithm against the adversary that answers 0 to every probe. Since the algorithm is deterministic, the probe sequence and final output are determined; let P be the set of probed pairs, and suppose $|P| \leq \binom{\lceil n/2 \rceil}{2} - 1$. The empty graph G_0 is consistent with all answers and has $\mu(G_0) = 0$. We claim the graph $H = K_n \setminus P$ (all unprobed pairs present) is also consistent — trivially, every probed pair is a non-edge of H — and satisfies $\mu(H) \geq n/4$. Suppose not: $\mu(H) < n/4$. Taking both endpoints of a maximum matching of H yields a vertex cover C of H with $|C| \leq 2\mu(H) < n/2$. Then $U = V \setminus C$ has $|U| > n/2$, hence $|U| \geq \lceil n/2 \rceil$ (integrality). No edge of H lies inside U (it would be uncovered), so *every* pair

inside U belongs to P , giving $|P| \geq \binom{|U|}{2} \geq \binom{\lceil n/2 \rceil}{2}$, a contradiction. So both G_0 ($\mu = 0$) and H ($\mu \geq n/4$) are consistent with the transcript, and whatever the algorithm outputs is wrong for one of them.

Randomized upper bound. Under the promise, if $\mu(G) \geq n/4$ then G contains $\geq n/4$ pairwise-disjoint edges, so a uniformly random pair is an edge with probability at least $(n/4)/\binom{n}{2} \geq 1/(2n)$. Probing $cn \log n$ independent uniform pairs finds an edge with probability $1 - n^{-\Omega(c)}$ if $\mu(G) \geq n/4$, and finds none if $\mu(G) = 0$; output accordingly. (Distinguishing only requires certifying $\mu \geq 1$ under the promise.)

Randomized lower bound. By Yao’s principle take the distribution: with probability $1/2$ the empty graph; with probability $1/2$ a uniformly random matching of size $n/4$ (on a uniformly random $n/2$ -subset). Any fixed probe is an edge with probability at most $(n/4) \cdot \frac{1}{\binom{n}{2}} \cdot \Theta(1) = O(1/n)$ under the second component; with $o(n)$ probes the transcript is all-zeros with probability $1 - o(1)$ under *both* components, forcing error $\geq 1/2 - o(1)$. $\square$

Remark 3.7. The bound $\binom{\lceil n/2 \rceil}{2} \approx n^2/8$ is tight up to the constant: probing all pairs costs $\binom{n}{2}$ and a slightly cleverer deterministic strategy can certify the gap after $\binom{n}{2} - \binom{\lceil n/2 \rceil}{2} + O(n)$ probes (*verify exact constant; minor*). The lemma is robust to making the promise bipartite, to replacing $n/4$ by δn , and to allowing the algorithm to output “a matching” instead of a bit. [S] for these routine variants.

Corollary 3.8 (Architectural barrier for derandomizing the polylog/ORS pipelines). *Fix any dynamic-matching algorithm whose update/query procedure invokes, as a black box, a matching-size gap estimator on an implicitly represented N -vertex graph accessed via adjacency probes — as do [BKS23, Beh23, ABR24, BG24, AKK25] via RGMM-style sublinear estimation [Beh21] (Section 2.4). If the black box is replaced by any deterministic probe algorithm with the same interface and guarantee, then by Lemma 3.6 some calls cost $\Omega(N^2)$ probes in the worst case, destroying the sublinear per-call budget that the frameworks’ analyses rely on. Hence there is no deterministic drop-in replacement at the oracle layer: derandomizing these pipelines requires re-architecting them around incrementally maintained deterministic summaries rather than fresh per-call sublinear estimation. [P] (as a statement about the probe interface; it is not an unconditional lower bound for OP-D, and we will say so wherever it is used.)*

Remark 3.9 (Pseudorandomness does not contradict the barrier). A “deterministic” algorithm that fixes a PRG seed is still a deterministic algorithm, and Lemma 3.6 applies: the worst-case adversary may depend on the fixed seed. This cleanly explains why naive “just plug in Nisan/ k -wise ranks” plans for RGMM fail at the model level, while leaving open the legitimate route of *maintained* structures (WS-A3). [P]

3.5 Consistency checks that calibrate difficulty [P]

Observation 3.10 (D2 implies beating the randomized frontier). A deterministic polylog-time $(1+\varepsilon)$ -approximate *fractional* matching (target D2), combined with deterministic rounding [BK21, ACCSW18], would yield an integral $\approx \frac{2}{3}(1 - O(\varepsilon))$ -approximation in polylog time (the $3/2$ integrality gap of the matching LP on general graphs is the bottleneck), i.e. ratio $\approx 0.66 > 2 - \sqrt{2} \approx 0.586$ — which would improve the best *randomized* polylog ratio [ABR24]. Hence D2 is at least as hard as a major open randomized question; the plan treats D2 as a stretch target and orders work accordingly. The same calibration shows polylog-time EDCS maintenance is open even randomized (it would likewise beat [ABR24]). [P] (modulo the exact rounding loss; verify in Task 14(d).)

Observation 3.11 (What OP-L can and cannot mean, honestly). “Polynomial communication lower bounds for deterministic dynamic algorithms,” read literally as unconditional polynomial cell-probe bounds, is beyond current technology regardless of determinism (Section 2.5); and in bounded-space dynamic streams the randomized bounds [AKLY16] are already near-maximal, leaving no det/rand daylight. The defensible interpretations are exactly L1–L5 of Problem 3.2; the plan commits to L1+L2 unconditionally (deliverable in months), pursues L3/L5 as research bets, and uses L4 as the bridge to Workstream B. [P] (as an assessment, with the cited evidence.)

4 Workstream A: derandomization (OP-D)

Guiding principle from Corollary 3.8: do not derandomize the oracle calls; *remove* them. Every task below replaces “fresh sublinear estimation” by “maintained deterministic structure” somewhere in the stack.

Task 1 (A0: randomness audit, formal version). Produce a table mapping every random choice in [BKS23, Beh23, ABR24, BG24, AKK25] to (i) the property it buys (correctness whp / adaptivity / amortization), (ii) the failure mode if fixed, (iii) whether limited independence suffices in the analysis. Output: 4–6 page technical note; feeds A1–A4 and is reusable as a related-work section. Effort: 2–3 weeks. Risk: low.

Task 2 (A1: deterministic fractional route, graded). Current det. frontier: $(2 + \varepsilon)$ -FRAC in $O(1)$ [BCH17]; det. rounding exists [BK21]. Subgoals, in order: (a) det. $(2 - \delta_0)$ -FRAC in polylog for some explicit $\delta_0 > 0$ via primal-dual with deterministic level resets ([BHI15, BHN16] machinery); (b) understand precisely why the [BHN16] better-than-2 bipartite algorithm pays n^γ — is the bottleneck an estimation primitive (then Corollary 3.8 predicts an obstruction) or amortized rebuilding (then deterministic multi-scale rebuilding may help)? (c) only then attempt D2, knowing Observation 3.10 prices it. Effort: 2–3 months for (a)–(b). Risk: medium.

Task 3 (A2: EDCS route). Edge-degree-constrained subgraphs give $(3/2 + \varepsilon)$ guarantees and are maintainable deterministically in $O(m^{1/4})$ -type worst-case time [GSSU22, BS16]. Subgoal: deterministic EDCS maintenance in $n^{1/2-\delta}$, attacking the rebuild bottleneck with deterministic expander-decomposition-style partitioning (as in [BBKS25], whose $n^{8/9}$ comes from exactly such tools). Calibration: full polylog EDCS is open even randomized (Observation 3.10), so the target is the exponent, not polylog. Effort: 3 months. Risk: medium-high.

Task 4 (A3: the crux — rank-robust greedy / D1). Deterministic dynamic *maximal* matching in $n^{o(1)}$ (D1) is the keystone: it is the minimal primitive whose randomized version $O(1)$ [Sol16]) powers the kernels of the better-than-1/2 pipelines, and [BBKS25] just moved the det. bound to $\tilde{O}(n^{8/9})$. Attack plan: (a) reprove [BBKS25] from our audit template and isolate where 8/9 arises (their tradeoff between deterministic expander decompositions and rebuild schedules); (b) formalize the obstruction: for any *fixed* edge ranking π , a rank-aware adversary forces greedy-along- π to suffer $\Omega(n)$ recourse per update — prove this as a standalone lower bound for the natural “fixed-rank greedy” algorithm family [S]; (c) circumvent (b) with multi-scale rank hierarchies (Bentley–Saxe-style deterministic rebuilding trading recourse for a $(1 - 1/\text{polylog})$ approximation slack), targeting det. *almost*-maximal $((1/2 - o(1))$ -approx) in $n^{3/4}$, then $n^{1/2+o(1)}$. Effort: the core 4–5 months of WS-A. Risk: high, with publishable intermediates at every exponent improvement and at the lower bound (b).

Task 5 (A4: limits of pseudorandomness, short note). Determine the minimal independence/seed length under which the RGMM locality analyses [Beh21] survive. *Verified obstacle (v1.1)* [V]: Beh21’s Lemma 3.10 charges queries via the map $\phi(\pi, \vec{P})$ that rotates ranks along query paths — a bijection of the *full* symmetric group; k -wise families are not closed under these rotations, so the $O(\bar{d} \log n)$ analysis does not transfer to limited independence as-is, while experiment E3 of `audit_lab` finds even 2-wise ranks empirically matching full independence on oblivious instances — a genuine analysis/experiment gap that this task should close in either direction. Combine with the Remark after Corollary 3.8 into a note “what pseudorandomness can and cannot do for dynamic matching.” Effort: 3–4 weeks, parallelizable. Risk: low (the note is valuable even if the answer is “full independence needed”).

Task 6 (A5: deterministic ORS-parameterized algorithm (D4)). Only after A3 yields a det. kernel primitive: re-derive the [AKK25] epoch structure with deterministic peeling, replacing whp-union-bounds over epochs by potential functions. The honest expectation [C] is det. $\text{EDGES}(\varepsilon)$ in $\tilde{O}(\sqrt{n \cdot \text{ORS}(n)})$ -type time first (the [BG24] shape), since [AKK25]’s improvement leans hardest on random sampling. *Refined target (v1.1)* [V]: by the A0 audit, D4 reduces to derandomizing three modules; the only one not covered by Lemma 3.6 is DET-OPPORTUNISTIC-MATCHING — given adjacency access and the *promise* $\mu(G[U]) \geq \delta n$, deterministically output a $\gamma \delta n$ -matching in $\tilde{O}(m n^{O(\gamma)} / \Delta_{\text{IN}}(M))$ time (AKK25 Lemma 3.1 shape; $\tilde{O}(n^{2+O(\gamma)} / d_{\text{avg}})$ for the BG24 Lemma 7 shape). This is a *promise-search* problem, outside the decision lower bound of Lemma 3.6; the worst case $\Omega(m)$ is necessary even randomized [2], so only the Δ_{IN} -adaptivity needs replication.

5 Workstream B: the combinatorics of ORS/RS density

By [AKK25]+[Pra26], *any* progress on RS density at linear matching size transfers to the dynamic-matching frontier in both directions; this workstream is the highest-variance, highest-reward component, and also produces the experimental deliverables.

Task 7 (B0: the (r, t) frontier table). Compile, with verbatim verified statements, the best known bounds on $\text{RS}(n, r)$ and $\text{ORS}(n, r)$ across regimes: $r = n/2 - o(n)$ (constant t regimes), $r = \delta n$ (the [FLNRRS02] $n^{1/\Theta(\log \log n)}$ construction; the exact δ -dependence used by [AKK25]), $r = n^{1-o(1)}$ (Behrend regime, $t = n^{1-o(1)}$ [Beh46, RS78]), including corner-free-set improvements [CFTZ22]. Identify *the* regime that controls the [AKK25] runtime (their ε -dependence) — this is the single most important verification in the plan. Output: survey section + machine-readable table. Effort: 3–4 weeks. Risk: low.

Task 8 (B1: `ors_lab` — exact small- n values). Build a verified search package computing exact $\text{ORS}_r(n)$ and $\text{RS}_r(n)$ for $n \lesssim 40$: (a) two *independent* encodings (SAT via incremental CNF with symmetry breaking; CP-SAT/ILP) cross-validated on all instances, per the standing methodology of this research program; (b) certificates: witness colourings checked by a separate $O(t \cdot n^2)$ verifier; UNSAT instances logged with solver proofs where available; (c) outputs: exact tables, extremal examples, and a targeted search for small- n gaps $\text{ORS}_r(n) > \text{RS}_r(n)$ — Pratt’s equivalence is asymptotic, so small- n separations are possible and would inform constructions (and make a nice figure regardless of sign). [C] on finding separations. Effort: 4–6 weeks including writeup. Risk: low (the tables are a contribution per se; nearest analogue: exact small- n extremal tables in the induced-matchings literature).

Task 9 (B2: constructions). Lift recent corner-free/cap-set technology [CFTZ22, BCCGNSU17] into the $r = \delta n$ regime: the concrete question is whether combinatorial degenerations improve the

$n^{1/\Theta(\log \log n)}$ of [FLNRRS02] for *any* fixed $\delta > 0$. Even a log-factor-in-the-exponent improvement directly worsens the conditional optimality picture for [AKK25] and is a standalone combinatorics paper. [C]. Effort: open-ended; timebox 2 months with B1’s extremal examples as seeds. Risk: high.

Task 10 (B3: upper bounds where ordering should not help). The classical upper bounds for $\text{RS}(n, \delta n)$ go through triangle-removal/regularity; the ordered variant breaks the symmetry those proofs use, which is exactly why [BG24] had to ask the ORS question separately — and [Pra26] now supplies the reduction $\text{ORS} \rightarrow \text{RS}$, so the leverage point is: prove $\text{RS}(n, \delta n) = n^{o(1)}$ for δ below the current threshold, or a conditional/regime-restricted version, then transfer. Any such bound *upper-bounds the runtime of [AKK25]* (an L4-type result: it makes the algorithm unconditionally fast, the optimistic resolution of the program). Effort: research bet; timebox 6 weeks initial. Risk: high. [C]

6 Workstream C: deterministic-specific lower bounds (OP-L)

Task 11 (C1: probe-model separations, paper-grade (L1)). (a) Write up Lemma 3.6 with the variants (bipartite, δn gaps, output-a-matching), the tightness remark, and the [ABKRS04] comparison. *Prior-art scoping (v1.1)* [V]: Assadi–Solomon [1] (Thm. 3) already prove a deterministic $\Omega(n^2)$ vs. randomized $\tilde{O}(n\beta)$ separation for *finding* a maximal matching on adjacency-array graphs of neighborhood independence $\beta = 2$, and [2] (Thm. 6) shows finding a maximal matching needs $\Omega(n^2)$ queries even randomized. The note’s novelty claim must therefore rest on: the adjacency-matrix probe model, the size-estimation (decision) task — where randomized is $\tilde{O}(n)$ [Beh21], unlike finding — the explicit clique-coverage threshold $\binom{n/2}{2}$, and the pipeline barrier of Corollary 3.8; cite both works prominently. (b) *List/adjacency-array model*: the matrix-model adversary is too weak there (degrees reveal emptiness in $O(n)$), so plant the gap on top of a fixed background graph: target — any deterministic algorithm achieving $\text{SIZE}(1/2 + \delta)$ needs $n^{1+\Omega(1)}$ (stretch: $\Omega_\delta(n^2)$) adjacency-list probes; method — derandomized congestion adversaries adapted from the dummy-vertex machinery of [BRR24], replacing their distributional core by an explicit potential the adversary maximizes against the fixed algorithm. [S] (c) *Exhaustive small- n verification*: implement the adversary and verify the exact bound $\binom{n/2}{2}$ for $n \leq 12$ by exhaustive simulation (ties into WS-D). Deliverable: a SOSA-style note “Deterministic probe complexity of approximate matching size” = L1+L2 packaged (Lemma 3.6 + Corollary 3.8 + (b) if it lands). Effort: (a),(c) 3–4 weeks; (b) 2 months. Risk: low/medium.

Task 12 (C2: dynamic permutation hiding under bounded recourse (L5)). Observation 3.11 forces a resource restriction. Proposed model: *bounded-recourse deterministic dynamic algorithms* — the algorithm maintains an explicit matching and may change $\leq \rho(n)$ edges per update; lower-bound ρ for approximation $1 - \varepsilon$ via a k -phase adaptive game in which the adversary inserts the layers of a permutation-hiding ORS gadget [AS23] in an order chosen *adaptively against the deterministic transcript* (this is the step Yao-based proofs cannot do, hence the deterministic-specific hope). Milestones: (a) formal model + trivial bounds; (b) recourse lower bound $\rho = \Omega(\text{poly}(\text{ORS-density}))$ for fixed-rank greedy (connects to Task 4(b)); (c) general deterministic bounded-recourse bound [X]. Honest wall, stated upfront: recourse bounds do not bound update *time* for SIZE algorithms; they target EDGES. Effort: 3 months. Risk: high.

Task 13 (C3: a Det-OMv hypothesis (L3)). (a) Formulate: “no deterministic algorithm solves OMv in $n^{3-\delta}$ total time (with polynomial preprocessing)” and the ε -approximate variants matching [Liu24]; first sanity step is a focused search for deterministic subcubic(-by-polynomial-factor)

OMv — note [LW17] is randomized, and the deterministic status of even $n^3/2^{\Omega(\sqrt{\log n})}$ must be pinned down (Task 14(e)). (b) Check the [HKNS15, Liu24] reductions are deterministic (they are reduction-side deterministic on inspection [S]), so Det-OMv transfers to deterministic dynamic lower bounds with the same quantitative shape. (c) Gadget program: strengthen specific reductions using *deterministic* communication primitives (Equality/Greater-Than have $\Theta(n)$ deterministic but $O(\log n)/O(1)$ randomized communication) to get deterministic-only separations in intermediate communication-game models. [C] Effort: (a),(b) 1 month; (c) open-ended, timebox 2 months. Risk: medium — the failure mode “hypothesis is false” is itself a (smaller) result.

7 Workstream D: engineering, verification, packaging

Per the standing methodology of this research program (every paper ships runnable, cross-validated code):

- **ors_lab/**: Task 8; SAT (PySAT/kissat) and CP-SAT encoders, certificate verifier, table generator, plots.
- **probe_lb/**: Task 11(c); adversary implementation, exhaustive verification harness $n \leq 12$, and a fuzz harness pitting published randomized estimators against the adversary instances (illustrative figures for the note).
- **audit/**: machine-readable randomness-audit table (Task 1) rendered to LaTeX.
- Reproducibility: pinned environments, seeds, CI on small instances; independent re-implementation rule for any number that enters a paper.

Effort: woven into B1/C1; ~ 3 weeks dedicated. Risk: low.

8 Phase 0 verification, timeline, gates, fallbacks

Task 14 (V.1 (Phase 0): verification pass — blocking). Before any proof is trusted: (a) exact [AKK25] theorem statements (ε -dependence, the precise ORS regime $r(\varepsilon)$, adaptive guarantee); (b) exact [Pra26] parameter translation; (c) exact [Liu24] equivalence quantifiers (value vs. witness vs. edges; bipartite only?) for Observation 3.4; (d) exact rounding losses in [BK21, ACCSW18] for Observation 3.10; (e) deterministic status of subcubic-savings OMv (for Task 13); (f) re-check all [M]-tagged citations (page numbers, venues); (g) arXiv sweep for 2025–26 successors of [BBKS25, ABR24, AKK25]. Output: corrected Section 2 + errata list. Effort: 2–3 weeks.

8.1 Timeline (9–12 months) and decision gates

Phase	Months	Content and exit gate
M0	0–1	Task 14; A0 audit started. <i>Gate G0</i> : premises confirmed; else re-plan.
M1	1–3	C1(a,c) note drafted; B1 ors_lab v1 ($n \leq 30$); A0 done. <i>Gate G1</i> : if C1(b) resists 4 focused weeks, pivot C1(b)→C2 recourse model.
M2	3–6	A3 (a,b): [BBKS25] reproved, fixed-rank recourse LB; B2 timebox. <i>Gate G2</i> : exponent progress on D1? If yes, A3(c) becomes the main paper track.
M3	6–9	A1/A2 in parallel with C3(a,b); B3 timebox. <i>Gate G3</i> : pick main paper.
M4	9–12	Integration, writing, code release; submit main + note(s).

8.2 Fallback publishable units (pre-committed)

1. *SOSA/short*: “Deterministic probe complexity of approximate matching size and a barrier for derandomizing dynamic matching” (Lemma 3.6 + Corollary 3.8 + variants) — deliverable from M1 material alone.
2. *Combinatorics/experimental note*: exact small- n ORS/RS tables with certificates and any ordered-vs-unordered small- n separation (B1).
3. *Position/SoK only if both stall*: “What would it take to derandomize dynamic matching?” — the audit + no-go observations + honest taxonomy; lowest value, explicitly last resort.

9 Risk register

Risk	Likelihood / impact	Mitigation
ORS/RS density resolved by others (either direction)	low / very high	Monthly arXiv monitor (combinatorics + data structures); plan survives both outcomes: sparse $\Rightarrow$ B3/L4 framing, dense $\Rightarrow$ constructions paper.
C1(b) list-model LB harder than hoped	medium / medium	Pre-planned pivot at G1 to the recourse model (C2), which reuses the machinery.
Det-OMv false (det. subcubic-savings OMv exists)	medium / medium	Sanity-search first (C3(a)); a refutation is itself a small result and redirects C3(c).
[BBKS25] techniques extended below $n^{8/9}$ during the project	medium / medium	A3(a) keeps us conversant; exponent race has many rungs; recourse LB A3(b)/C2 is differentiated.
A [M] citation or a premise fails verification	low / high	Phase 0 is blocking by design; claim-status appendix localizes damage.
Scope creep into the randomized-open zone (ratios $> 2 - \sqrt{2}$)	medium / low	Observation 3.3 consulted at every gate; such targets require explicit re-scoping.

A Randomness audit table (Task A0, v0.3)

Auto-generated from `audit_lab/auditlib/audit_table.py`; regenerate with `run_all.py`. Evidence tags E1–E5 refer to the experiments of the `audit_lab` package; see the A0 technical note for per-row discussion.

Table 1: Randomness audit (Task A0, v0.2). Tags: [V] verified against source, [M] internal reference pending, [P] proved in the plan, [E:*] empirical evidence in this package.

Id	Mechanism (papers)	(i) Buys	(ii) Failure if fixed	(iii) Limited in-dep.	Evidence
R1	Sublinear Monte-Carlo matching-size estimation via adjacency-matrix queries (Beh21). Verified statements: BG24 Prop. 4; AKK25 Prop. 2.3 (Otilde(n) poly(1/eps)) queries, whp $0.5\mu - \epsilon n \leq \text{est} \leq \mu$; BKSW23 abstract ('invoke ... Behnezhad (FOCS'21) in a white-box manner' for the second streaming pass); Beh23 SS1 ('key tool ... estimating the size of maximal matchings in Otilde(n) time'); ABR24 SS1 (phase 2 'augment ... by employing the sublinear-time matching size estimator of [Beh21]'). (all five (engine: Beh21))	Per-rebuild Otilde(n) cost with whp additive accuracy; the only access to the (kernel/contracted/leftover) graph is via probes.	Adversary who knows the fixed probe/sample set plants all matching mass outside it: estimator reports ~ 0 while $\mu \geq n/4$. Provably no deterministic probe algorithm with $o(n^2)$ probes exists (plan Lemma 3.6); no drop-in derandomisation of this interface (plan Cor. 3.7).	Chebyshev-level accuracy needs only pairwise sample choices; whp needs $k = \Theta(\log n)$ -wise tails. Whether the surrounding union bounds tolerate the weaker tails is Task A4. Beh21-internal requirements not yet re-read.	E1 (works); E5 (fixed fails); Lemma 3.6; Cor 3.7 [V][P][E:e1,e5]

Id	Mechanism (papers)	(i) Buys	(ii) Failure if fixed	(iii) Limited in- dep.	Evidence
R2	Uniformly random edge/vertex ranks driving greedy maximal matching: (a) inside the Beh21 estimator (local RGMM oracle); (b) as the explicitly maintained phase-1 maximal matching of the size pipelines – ABR24 SS1: 'maintain a maximal matching explicitly using the fast algorithms of [BGS, Sol16, BDHSS19]' (randomized, oblivious-adversary maintenance). (Beh21 inside all five; phase-1 of Beh23/ABR24)	Verified (Beh21 Thm 3.5): $E[T(v, \pi)] = O(\bar{d} \log n)$ over uniformly random v AND uniformly random π , via the directional oracles VO/EO (Beh21 Algs 1-2, Remark 3.3); caching gives $\text{poly}(n)$ worst case with prob. 1 (Remark 3.2, Obs 3.7). In ABR24 the random order is ADDITIONALLY load-bearing for the approximation analysis of the b-matching (even-spread-of-copies argument, ABR24 SS2), not only for locality.	Rank-aware adversary builds a rank-decreasing path: one oracle query costs $\Theta(m)$ (E2); head-edge toggling forces $\Theta(n)$ recourse per update (E4). Approximation is NOT what breaks (any maximal matching stays $1/2$).	VERIFIED OB-STRUCTION (Beh21 SS3): the charging proof of Lemma 3.10 maps a permutation π to $\phi(\pi, P)$ by rotating ranks along query paths – a bijection of the FULL symmetric group. k -wise families are not closed under these rotations, so the $O(\bar{d} \log n)$ locality analysis does not transfer to limited independence as-is (Task A4 now has a precise technical obstacle). Empirically (E3): even $k=2$ matches full independence on oblivious instances – a genuine analysis/experiment gap. BDHSS19 maintenance internals unread [M].	E2 ($\Theta(m)$ blowup); E4 ($\Theta(n)$ recourse); E3 (k -wise) [V][M][E:e2,e3,e4]

Id	Mechanism (papers)	(i) Buys	(ii) Failure if fixed	(iii) Limited in- dep.	Evidence
R3	Adaptive-adversary robustness wrappers. Verified claims: BKS23 abstract (adaptive + worst-case polylog, both whp); BG24 SS3 (adversary sees previous outputs) + Thm 1; AKK25 Result 1 + Problem 1 (adaptively chosen queries); Beh23 SS1 (vertex sparsification 'can be made work against adaptive adversaries too [Kis22]'); ABR24 Thm 1 ('poly(log n) worst-case update time even against adaptive adversaries', via LOCAL-model simulation on the bounded-degree $G[M \cup B]$ + size-only output); BKS23 worst-case via deamortization (JACM SS2: 'easily deamortized'). Mechanism: low-sensitivity outputs (sizes; or an explicit matching only shrunk by deletions between rebuilds, BG24 Lemma 9) + fresh coins per rebuild/call + whp union bounds over poly(n) updates. (BKS23; Beh23; ABR24; BG24; AKK25)	Correctness whp against adversaries that see the outputs.	With fixed coins the transcript determines internal randomness; every adversary is effectively adaptive against a deterministic algorithm – OP-D has no 'oblivious deterministic' easy mode (plan SS2.1, Obs 3.4).	Across rebuilds: coins must be fresh/independent. Within a rebuild: inherits from R1/R2/R4/R5.	conceptual; plan SS2.1, Obs. 3.4; BG24 Alg 2 ll.17-20 [V][M]
R4	RandomSampling (BG24 Algorithm 1, Lemma 7): per-vertex uniform sampling of candidate partners with geometrically doubling budgets $b = n^{\{2\delta(i-1)\}}$; whp degree bound via Claim 8 (per-(v,i) failure $1/n^3$, union bound). Fresh coins per call; called from MatchAndCertify (Lemma 6) inside Algorithm 2 every eps $n/2$ updates. (BG24)	Finds an $\Omega_{\text{eps}}(n)$ matching in $G[U]$ in $\tilde{O}(n^{\{2+\text{eps}\}}/d)$ where $d = \text{avg internal degree of the certificate}$ – fast exactly when the instance is far from induced.	Adversary aligns the instance with the fixed sample schedule: planting off-the-samples (E5 mechanism) breaks the Claim-8 degree invariant. NOTE (verified): the ORS charging itself (Lemma 11), H_{cert} bookkeeping, and the boosting wrapper (Prop. 3) are deterministic – the coins of BG24 live ONLY in Lemma 7 + the R1/R7 interfaces.	Claim 8 needs $(1-p)^{\{\Omega(\log n / p)\}} \leq n^{\{-\Theta(1)\}}$ tails: $\Theta(\log n)$ -wise independence plausibly suffices (Bonferroni truncation) – unproven, input to Task A4 [S].	E5 (mechanism); BG24 Lem 7, Claim 8, Lem 11, Alg 2 [V][E:e5]

Id	Mechanism (papers)	(i) Buys	(ii) Failure if fixed	(iii) Limited in- dep.	Evidence
R5	Opportunistic sublinear matcher (AKK25 Algorithm 1, Lemma 3.1): per-vertex neighbour sampling with geometric probabilities $p_i = n^{\{3 \gamma_i\}}/n$ (binomial-count implementation, fn.1); whp runtime via Claim 3.3 (Chernoff) and max-internal-degree via Claim 3.4 ('residual sparsity' + union bound). Used inside the three-graph batch scheme ($G_{\text{old}}/G_{\text{batch}}/G_{\text{match}}$), - FINDING a Alg 2, Lemma 4.1) and recursed via the family $\{A_i\}$ (SS4.2). (AKK25)	Strengthens BG24 Lemma 7: $O(\tilde{m} n^{\{3 \gamma\}}/\Delta_{\text{IN}}(M))$ time, sensitive to $ E $ and to MAX (not avg) internal degree - this is what removes the $\sqrt{n}$ factor. Worst-case $\Omega(m)$ is necessary: ACK19 (Assadi-Chen-Khanna SODA'19) Thm 6, - FINDING a maximal matching needs $\Omega(n^2)$ queries even for randomized algorithms.	Same plant-off-the-samples class as R4; the recursion compounds the exposure (each level reruns the sampler). Deterministic route expectation unchanged: BG24 shape first (plan Task A5, [C]).	Claims 3.3/3.4 are Chernoff + union bound: $\Theta(\log n)$ -wise plausibly suffices [S]; recursion multiplies the number of seeds, not their required strength. SS4.2/SS5 details read at overview level only.	AKK25 Lem 3.1, Claims 3.2-3.4, Lem 4.1; E5 (mechanism) [V][M]
R6	Whp amplification: union bounds over $\text{poly}(n)$ updates / calls (explicit in BG24 Claim 8 and AKK25 Claims 3.3-3.4); fresh repetition where a single run succeeds with constant probability. (all five)	High-probability correctness, union-boundable over time.	No deterministic analogue inside the probe interface (plan Lemma 3.6 kills the single run, so repetition cannot help); outside it, replacement = maintained deterministic summaries (plan WS-A).	Repetitions must be mutually independent.	Lemma 3.6; Cor 3.7 [P]

Id	Mechanism (papers)	(i) Buys	(ii) Failure if fixed	(iii) Limited in- dep.	Evidence
R7	Randomized vertex sparsification for the additive-to-multiplicative reduction: contract vertices into $O(\mu/\epsilon)$ groups preserving $(1-\epsilon)$ of the matching, whp [AKLY16, AKL16, CCE+16] as implemented dynamically by Kis22. Verified statements: BKS23 Prop. A.1 (App. A): 'a randomized algorithm which for each update to G makes an update to $O(\log^2 n / \epsilon^3)$ contracted subgraphs, such that w.h.p. ...'; AKK25 Prop. 2.4; BG24 Prop. 5; Beh23 SS1 (adaptive via [Kis22]). (Beh23; BG24 (Prop. 5); AKK25 (Prop. 2.4); via Kis22)	Reduces vertex count to $O(\mu/\epsilon)$, turning ϵ n-additive guarantees into $(1-\epsilon)$ -multiplicative with polylog overhead, whp, including against adaptive adversaries.	A fixed (published) contraction partition lets the adversary place the matching across collision classes, destroying preservation – same plant-against-the-known-seed pattern as E5, one level up.	Preservation proofs bound per-vertex collision events; pairwise independence gives the expectation, whp needs stronger tails. Kis22 full internals unread [M] (its operative form, BKS23 Prop. A.1, is verified).	E5 (pattern); read-through pending for Kis22 lemmas [V][M]

Id	Mechanism (papers)	(i) Buys	(ii) Failure if fixed	(iii) Limited in- dep.	Evidence
R8	ALREADY-DETERMINISTIC layers, verified: McGregor boosting derandomized by Tir18 (explicit in AKK25 Prop. 2.2); greedy matching (AKK25 Fact 2.1, BG24 GreedyMatching); ORS charging (BG24 Lemma 11; AKK25 Lemma 2.5 – its probabilistic method lives in the ANALYSIS only, no algorithmic coins); H_cert / G_old-G_batch-G_match bookkeeping (BG24 Alg 2; AKK25 Alg 2); Gupta-Peng bounded-degree (1+eps) maintenance (deterministic, cited e.g. BLM20 Lemma 6.1); the Gupta-Peng periodic-recomputation paradigm itself (BKSW23 JACM SS2); and – notably – ABR24 Thm 2: a DETERMINISTIC two-pass semi-streaming (2-sqrt2-eps)-approximation, showing the combinatorial core (maximal matching + duplicate-free b-matching + blossom-inequality analysis) is already deterministic under pass access; the dynamic version’s coins are forced purely by the probe access model. (inventory row (not a randomness site))	– (no coins; listed to delimit the derandomization frontier)	n/a	n/a	Frontier consequence: derandomizing BG24/AKK25 reduces to exactly three modules: the opportunistic sampler (R4/R5), the Beh21 estimator interface (R1; provably no probe drop-in), and vertex sparsification (R7). The sampler is a SEARCH problem with a promise – outside Lemma 3.6’s decision lower bound – hence the genuinely open algorithmic target (refined Task A5). [V]

B Claim-status appendix

Tag semantics: [V] verified against the published source/arXiv in the June 2026 pass; [M] canonical memory citation, re-check before submission; [P] proven in this document; [S] proof sketch, completion is a named task; [C] conjecture/target; [X] speculation.

Claim	Status	Where
[BG24]: $(1 - \varepsilon)$ -EDGES, $O(\sqrt{n^{1+O(\varepsilon)}} \text{ORS}(n))$, whp, adaptive	[V]	§1.2
[AKK25]: same guarantee, $n^{o(1)} \text{ORS}(n)$; improvement is the <i>up-date time</i>	[V]	§1.2
[Pra26]: polynomial ORS $\Rightarrow$ polynomial RS (linear r)	[V]	§1.3
[BBKS25]: det. maximal in $\tilde{O}(n^{8/9})$; rand. adaptive $\tilde{O}(n^{3/4})$	[V]	§1.3
[ABR24]: $\text{SIZE}(2 - \sqrt{2} - \varepsilon)$ in polylog, general graphs	[V]	§2.3
[Liu24]: OMv equivalences; witness-vs-value distinction	[V]	§2.5
[AS23]: pass LBs via permutation hiding on RS graphs	[V]	§2.5
[BRR24]: randomized $\Omega(n^{2-\delta})$ list-queries for additive εn	[V]	§2.5
(r, t) frontier in the near- $n/2$ regime; [ABKL23] pages; [AKLY16, BGS11, BK21, GSSU22] details; cell-probe refs	[M]	Task 14(f)
Obs. 3.3 (no-go for randomized-robust techniques)	[P]	§3
Lemma 3.6 (quadratic det. probe bound; $\Theta(n^2)$ vs $\tilde{\Theta}(n)$)	[P]	§3.4
Cor. 3.8 (no det. drop-in at the oracle layer; framework-relative)	[P]	§3.4
Obs. 3.10 (D2 $\Rightarrow$ beating [ABR24]; modulo exact rounding loss)	[P]/[S]	§3.5
Obs. 3.4 (ORS sparse $\Rightarrow$ dyn. approx. OMv)	[S]	Task 14(c)
Fixed-rank greedy recourse LB; list-model det. LB	[S]	Tasks 4(b), 11(b)
Small- n ORS $>$ RS separations; B2/B3 outcomes; C2(c); C3(c)	[C]/[X]	WS-B, WS-C

C Notation

n, m	vertices / edges of the dynamic graph
$\mu(G)$	maximum matching size
$\text{RS}(n, r)$, $\text{ORS}(n, r)$	max # of (ordered-)induced edge-disjoint size- r matchings (Def. 2.1)
$\text{RS}(n)$, $\text{ORS}(n)$	the $r = \Theta(n)$ regime, as in [BG24]
EDGES/SIZE/FAC/WITNESS-VC	maintenance objectives (§2.1)
OMv	online matrix-vector multiplication (conjecture of [HKNS15])
whp / adp / obl	with high probability / adaptive- / oblivious-adversary
$\tilde{O}(\cdot)$	suppresses polylog(n) factors
D1–D4, L1–L5	sub-targets of OP-D / answer types of OP-L (§3)

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